1. **D**. The direction of **v** may change if **a** is not along **v**; the magnitude of **v** (the speed v) may not be changing uniformly.
2. **B**. Block A has only one degree of freedom (moving vertically), while block B has two (N=3, l=1, then s=3-1=2, swinging left and right, or back and forth). The total is 3.
3. **C**. (A) $U = -GMm/r$. (B) The integral of a one-variable function is always invariant in an arbitrary interval, because the path is unique. But you must be sure that it does not vary with time, otherwise it becomes a function of both x and t. (C) It should be the square of amplitude. (D) $\omega = \sqrt{k/m}$.
4. **C**. (A) The torque does no work if it is always perpendicular to the rotation. (B) Moment of inertia is determined by both the mass and the distribution around the axis. (C) Kinetic energy of rotation: $K = L^2/2I$. (D) The acceleration of a part includes tangential acceleration and centripetal acceleration (向心加速度). The former is proportional to the angular acceleration of the whole rigid body, while the latter is not.
5. **D**. (A) The curl of electrostatic field is always zero, while the divergence of induced electric field is always zero. (B) A tangential component of electric field on the surface generates electric current, and it is not an electrostatic equilibrium. (C) $\mathbf{E}\cdot \mathrm{d}\mathbf{A} = 0$ if **E** is along d**A**. (D) EMF cannot be generated by "electrostatic interaction", not "electromagnetic interaction".
6. **A**. (A) $\mathbf{E}_\mathrm{i} = -\partial \mathbf{A}_\mathrm{M}/\partial t$. (B) The integral should be halved: $\mathbf{A}_\mathrm{M} = \frac{1}{2}\int \mathbf{B}\times \mathrm{d}\mathbf{r}$. (C) As a "potential", $\nabla\cdot\mathbf{A}_\mathrm{M}$ does not correspond to any physical field, so it can be altered. (D) As a "potential", it does not matter to add in an arbitrary constant.

   ※Proof of $\mathbf{A}_\mathrm{M} = \frac{1}{2}\int \mathbf{B}\times \mathrm{d}\mathbf{r}$: for $\mathrm{d}\mathbf{X} = \mathbf{B}\times \mathrm{d}\mathbf{r}$:

$$\mathrm{d}\mathbf{X} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ B_x & B_y & B_z \\ \mathrm{d}x & \mathrm{d}y & \mathrm{d}z \end{vmatrix} = \left(B_y\mathrm{d}z - B_z\mathrm{d}y\right)\mathbf{i} + \left(B_z\mathrm{d}x - B_x\mathrm{d}z\right)\mathbf{j} + \left(B_x\mathrm{d}y - B_y\mathrm{d}x\right)\mathbf{k}$$

$$\mathrm{d}X_x = B_y\mathrm{d}z - B_z\mathrm{d}y, \mathrm{d}X_y = B_z\mathrm{d}x - B_x\mathrm{d}z, \mathrm{d}X_z = B_x\mathrm{d}y - B_y\mathrm{d}x$$

   Hence

$$\begin{pmatrix} \frac{\partial X_x}{\partial x} & \frac{\partial X_x}{\partial y} & \frac{\partial X_x}{\partial z} \\ \frac{\partial X_y}{\partial x} & \frac{\partial X_y}{\partial y} & \frac{\partial X_y}{\partial z} \\ \frac{\partial X_z}{\partial x} & \frac{\partial X_z}{\partial y} & \frac{\partial X_z}{\partial z} \end{pmatrix} = \begin{pmatrix} 0 & -B_z & B_y \\ B_z & 0 & -B_x \\ -B_y & B_x & 0 \end{pmatrix}$$

$$\nabla\times\mathbf{X} = 2B_x\mathbf{i} + 2B_y\mathbf{j} + 2B_z\mathbf{k} = 2\mathbf{B}$$

   If we neglect all the items independent of the curl, there is $\mathbf{X} = 2\mathbf{A}_\mathrm{M}, \mathbf{A}_\mathrm{M} = \frac{1}{2}\int \mathbf{B}\times \mathrm{d}\mathbf{r}$.
7. The acceleration (in m/s^2):

$$\mathbf{a} = \frac{1}{2}\mathrm{e}^{-2t}\mathbf{i} - \frac{\sqrt{t}}{2}\mathbf{j} + \mathbf{k}$$

   The velocity (in m/s):

$$\mathbf{v} = \mathbf{v}_0 + \int_0^t \mathbf{a}\mathrm{d}t = \left(-2 + \frac{1}{4}(1 - \mathrm{e}^{-2t})\right)\mathbf{i} - \frac{1}{3}t^{\frac{3}{2}}\mathbf{j} + (1 + t)\mathbf{k}$$

$$= -\frac{1}{4}(7 + \mathrm{e}^{-2t})\mathbf{i} - \frac{1}{3}t^{\frac{3}{2}}\mathbf{j} + (1 + t)\mathbf{k}$$

The position (in meter):

$$\mathbf{r} = \mathbf{r}_0 + \int_0^t \mathbf{v}\mathrm{d}t = \frac{1}{8}(\mathrm{e}^{-2t} - 15)\mathbf{i} - \frac{2}{15}t^{\frac{5}{2}}\mathbf{j} + \left(t + \frac{1}{2}t^2\right)\mathbf{k}$$

(The numerical calculation is omitted.)

8. The equilibria satisfy:

$$F = 0, \frac{\pi}{x} = \left(k + \frac{1}{2}\right)\pi \;\rightarrow\; x = \frac{2}{2k+1}, k \in \mathbb{N}$$

In the interval 0.1<x<5, x=2, 2/3, 2/5, 2/7, 2/9, 2/11, 2/13, 2/15, 2/17, 2/19.

Stability:

$$\frac{\mathrm{d}^2U}{\mathrm{d}x^2} = -\frac{\mathrm{d}F}{\mathrm{d}x} = -\frac{\pi\sin(\pi/x)}{x^2}$$

x=2, 2/5, 2/9, 2/13, 2/17 are unstable; x=2/3, 2/7, 2/11, 2/15, 2/19 are stable.

9. Divide the block into coaxial thin rings: for radius $r = \sqrt{x^2 + y^2}$, there is

$$h = b - r^2 > 0 \;\rightarrow\; 0 < r < \sqrt{b}$$

$$\mathrm{d}V = 2\pi rh\mathrm{d}r = 2\pi(br - r^3)\mathrm{d}r, \mathrm{d}m = 2\pi\rho(br - r^3)\mathrm{d}r$$

Then

$$V = \int_0^{\sqrt{b}} 2\pi(br - r^3)\mathrm{d}r = \frac{1}{2}\pi b^2 \;\rightarrow\; \rho = \frac{m}{V} = \frac{2m}{\pi b^2}$$

Hence

$$\mathrm{d}m = \frac{4m}{b^2}(br - r^3)\mathrm{d}r, I = \int_0^{\sqrt{b}} r^2\mathrm{d}m = \int_0^{\sqrt{b}} \frac{4m}{b^2}(br^3 - r^5)\mathrm{d}r = \frac{1}{3}bm$$

10. (a) The kinetic energy:

$$K = \frac{1}{2}mv^2 = \frac{1}{2}mL^2\dot{\theta}^2$$

For small θ, there is $\sin\theta \approx \theta, \cos\theta \approx 1 - \frac{1}{2}\theta^2$ (Taylor series expansion), and gravitational & elastic potential energy:

$$U_1 = \frac{1}{2}mgL\theta^2, U_2 = \frac{1}{2}kL^2\theta^2$$

The Lagrangian function and general momentum:

$$L = K - U = \frac{1}{2}mL^2\dot{\theta}^2 - \frac{1}{2}(mgL + kL^2)\theta^2, p = \frac{\partial L}{\partial\dot{\theta}} = mL^2\dot{\theta}$$

The Hamiltonian canonical equations:

$$H = \frac{1}{2}mL^2\dot{\theta}^2 + \frac{1}{2}(mgL + kL^2)\theta^2, \dot{\theta} = \frac{\partial H}{\partial p} = \dot{\theta}$$

$$\dot{p} = mL^2\ddot{\theta} = -\frac{\partial H}{\partial\theta} = -(mgL + kL^2)\theta \text{ or } \ddot{\theta} + \left(\frac{g}{L} + \frac{k}{m}\right)\theta = 0$$

(b) If we cannot consider that $\sin\theta \approx \theta, \cos\theta \approx 1 - \frac{1}{2}\theta^2$, there is

$$U_1 = mgL(1-\cos\theta), U_2 = \frac{1}{2}kL^2\sin^2\theta, L = \frac{1}{2}mL^2\dot{\theta}^2 - mgL(1-\cos\theta) - \frac{1}{2}kL^2\sin^2\theta$$

$$H = \frac{1}{2}mL^2\dot{\theta}^2 + mgL(1-\cos\theta) + \frac{1}{2}kL^2\sin^2\theta, \dot{\theta} = \frac{\partial H}{\partial p} = \dot{\theta}$$

$$\dot{p} = mL^2\ddot{\theta} = -\frac{\partial H}{\partial \theta} = -mgL\sin\theta - kL^2\sin\theta\cos\theta \text{ or } \ddot{\theta} + \frac{g}{L}\sin\theta + \frac{k}{2m}\sin 2\theta = 0$$

11. (a) Gauss surface: coaxial cylinders of radius *r*, the electrostatic field is radian to the axis.

    For $r < R_1,\ 2\pi rhD = 2\pi rh\varepsilon E = \rho V = \rho\pi r^2 h,\ E = \rho r/2\varepsilon$

    For $R_1 < r < R_2,\ E = 0$

    For $r > R_2,\ 2\pi rh\varepsilon_0 E = \rho\pi R_1^2 h,\ E = \rho R_1^2/2\varepsilon_0 r$

    (b) The integral path: $r = 0$ to R_1 to R_2 to R_3

$$V = -\left(\int_0^{R_1}\frac{\rho r}{2\varepsilon}\mathrm{d}r + \int_{R_2}^{R_3}\frac{\rho R_1^2}{2\varepsilon_0 r}\mathrm{d}r\right) = -\frac{\rho}{4\varepsilon}R_1^2 - \frac{\rho R_1^2}{2\varepsilon_0}\ln\frac{R_3}{R_2}$$

12. Closed path: circles of radius *r* perpendicular to the axis, the magnetic field is circular around the axis.

    For $r < R_1,\ 2\pi rH = 2\pi rB/\mu_0 = jA = \pi r^2 I/\pi R_1^2, B = \mu_0 Ir/2\pi R_1^2$

    Note: the permeabilities of most materials can be regarded as μ_0, unless it is mentioned specifically.

    For $R_1 < r < R_2,\ 2\pi rB/\mu = I,\ B = \mu I/2\pi r$

    For $r > R_2$, the opposite electric currents cancel out, $B = 0$

13. (a)(b)(c) The electrostatic field, magnetic field and induced electric field:

$$\mathbf{E}_\mathrm{s} = -\nabla V = -2x\mathbf{i} + (2z-3)y\mathbf{j} + 2y\mathbf{k}$$

$$\mathbf{B} = \nabla\times\mathbf{A}_\mathrm{M} = 6yt\mathbf{i} - 2x\mathbf{k}$$

$$\mathbf{E}_\mathrm{i} = -\frac{\partial\mathbf{A}_\mathrm{M}}{\partial t} = -3y^2\mathbf{k}$$

    (d) The charge density with no medium:

$$\rho = \varepsilon_0\nabla\cdot\mathbf{E}_\mathrm{s} = -2\varepsilon_0$$

    (e) The conducting current density should be

$$\mathbf{J} = \nabla\times\mathbf{H} - \frac{\partial\mathbf{D}}{\partial t} = \frac{1}{\mu_0}(\nabla\times\mathbf{B}) - \varepsilon_0\frac{\partial}{\partial t}(\mathbf{E}_\mathrm{s} + \mathbf{E}_\mathrm{i})$$

    Neither of the electrostatic field nor the induced electric field is related to the time, so

$$\mathbf{J} = \frac{1}{\mu_0}(\nabla\times\mathbf{B}) = \frac{1}{\mu_0}(2\mathbf{j} - 6\mathbf{k})$$

14. The junctions:

$$I_1 = I_2 + I_3, I_4 = I_3 + I_9, I_5 = I_2 + I_4, I_9 = I_6 + I_8, I_5 = I_6 + I_7, I_7 = I_8 + I_1$$

    The loops:

$$-3I_2 = 2, 3I_1 - 2I_9 = 3, I_6 + 2I_9 = 2, -I_6 = 2$$

    (in amperes)